

Citation under Pressure: Deployment Constraints, Legal Citation Integrity, and the Limits of Deterministic Verification

Harsh DUBEY¹

New York University, New York, USA

Abstract. Courts now sanction lawyers for filing citations a language model invented, and most research responds by detecting such errors in finished briefs. This paper measures how they are produced, and what a checker does in the loop instead. Seven models from seven vendors, open-weight 30B to frontier, drafted argument sections for 48 U.S. Supreme Court matters under five conditions: baseline, a citation quota, a restriction to pre-1970 authority, a sanctions warning, and all three combined. Every citation in the 1,680 primary drafts was adjudicated deterministically, with no language-model judge: existence against an 18-million-row reporter index, quotations verbatim against opinion text, pinpoint pages against official reporter pagination, the checker validated on seeded faults and its residual error read by hand. Which failure appears differs across the seven. Under combined pressure the 30B model’s reporter-coordinate existence fell from 0.929 to 0.775, quotation accuracy fell for three of the seven, two under both prompt templates, and accurate quotations per draft fell for six of the seven on at least one template, two at the top of the range; Sonnet 5’s rate did not fall, most plausibly because it quoted less. Adjusting for the age of the cited opinion, which the date clause itself sets, removes about 45 percent of that clause’s effect and leaves a residual that a stratified check does not reproduce, so the design does not separate damaged quoting from a shift toward older opinions. Fed the checker’s findings, models satisfy it by deletion: of Sonnet 5’s 35 flagged quotations 3 were corrected and 30 deleted, and accurate quotations per draft rose for one model of seven. Carrying the cited passage in the flag raises repair, which with dequoting then overtakes deletion in three of the seven. Holding revision rounds fixed, revision with no feedback interpolates to a nominal precision of 0.56 to 0.66, or 0.37 to 0.52 under the two hand readings, below which a verifier in this loop destroys more correct work than leaving the model to revise alone. Both experiments make one point: a rate computed over what a model chooses to expose to a check can rise while the correct output behind it does not, so an evaluation should report the count beside the rate.

Keywords. legal citation, hallucination, large language models, verification, evaluation

1. Introduction

Since 2023, courts have repeatedly sanctioned lawyers who filed briefs with citations a language model invented, from *Mata v. Avianca* [1] to a docket a public tracker now

¹Corresponding Author: Harsh Dubey, New York University; hd2225@nyu.edu. Instruments, drafts, records, and ledger: <https://github.com/hardy30894/citation-under-pressure>.

counts in the thousands [2]. Most published work frames the problem as detection: given a brief that already contains errors, how many can a model find [3,4]? That is the right measure for reviewing a filed brief, not for asking which prompting conditions produce the errors, this paper’s question.

This paper asks two production-side questions. First, which prompting conditions cause citation failure, and for which models? Deployed prompts carry constraints of the kind Zhao et al. [5] studied for scholarly citations, where a publication-year window lowered citation existence even in capable models; we add a sanctions warning and know of no comparable study for law. Second, when a deterministic checker feeds its findings back into revision, does the model correct them, leave them, or change the draft so that the tests pass without the quotations being corrected?

Both questions need an instrument the field has said does not exist in public: the highest-recall published detection agent misses nearly half of incorrect pinpoint citations (52.8 percent recall), which its authors attribute to page-level information living behind commercial databases [3]. We built it from public sources: existence against a local index from CourtListener bulk data, verbatim quotation checking validated on seeded faults and audited on real drafts, and page-level pinpoint verification against the official pagination in the Caselaw Access Project’s archives [6], which mark page breaks in the federal reporters as well as the U.S. Reports.

Our primary setting is the closed-book use of general-purpose models, the setting of most sanctioned filings to date [2]; a grounded arm with ten retrieved authorities in the prompt measures what retrieval changes. Drafts under the four non-baseline conditions are the pressured drafts (1,344 here). No draft refuses, and the two that stop under 100 words are scored as they stand. The paper contributes a constraint-sensitive evaluation, in which reliability at a neutral prompt does not predict reliability under drafting constraints; deterministic verification of existence, quotation, and pinpoint page with no language-model judge, released with the drafts and with its residual error read by hand rather than assumed away; and the finding that verification is not repair, since flags fed back to the model raise measured accuracy by deletion and an imprecise verifier damages correct content.

2. Related Work

Legal citation reliability. Liu et al. [3] build a detection benchmark from real appellate briefs with injected errors and a taxonomy grounded in real court filings; we adopt their categories and measure the naturally occurring errors their limitations section asks for. LegalCiteBench [4] shows that models asked closed-book to supply the authorities a proposition rests on almost never produce the right citation, the best under 7 of 100; we ask whether the cases a model chooses exist. Dahl et al. [7] and Magesh et al. [8] measure hallucinated authority in chatbots and research tools.

Constraints and fabrication. Zhao et al. [5] is the closest design: five prompting regimes over scholarly claims, verified against two bibliographic services, which left 36 to 61 percent of citations unresolved. We transplant their year-window and breadth regimes to law against a full-text index that leaves 0.6 percent unresolved, and add a statistical model.

Revision with execution feedback. The loop follows execution-feedback work in code generation [9], whose random-feedback ablation is the analogue of our false arm; we add a no-feedback arm separating content from extra revision, and mixed arms.

AI and law. Blair-Stanek and Van Durme [10] found frontier models flipping the winning party across identical zero-temperature runs, which Section 4 bounds with repeated generations. Evaluation here pairs automated metrics with the checks lawyers perform [11,12]; citation integrity has public, machine-checkable opinions and coordinates, though binding a quotation to the authority a draft means is heuristic. Garneau et al. [13] found retrieving the paragraph a proposition rests on harder than retrieving the case, in EU material. Zhang et al. [14] show models miss overruled precedent, which our date clause makes more likely and our checker does not test; ConsRAG [15] constrains retrieval. Evaluations of generated legal argument [16,17] do not check the authority it cites.

3. Tasks, Conditions, and Instruments

Terms. In 347 U.S. 483, 495, the second page is the *pincite*, for a quotation the page holding the quoted language. For a draft d let A_d be its *adjudicable* citations, labelled exists or not found, and S_d its *scored* quotations, those it attributes to a case whose text the checker holds. The outcomes are

$$E_d = \frac{|\{c \in A_d : \text{exists}\}|}{|A_d|}, \quad R_d = \frac{|\{q \in S_d : \text{accurate}\}|}{|S_d|}, \quad K_d = |\{q \in S_d : \text{accurate}\}|,$$

reporter-coordinate existence, *strict quotation accuracy*, and the *count* of accurate quotations in the draft; the lenient rate R_d^ℓ counts near misses as accurate. A quotation the draft does not attribute is outside S_d and so outside R_d .

Matters. The matters are forty-eight Supreme Court cases from 1990 to 2020 drawn from the Supreme Court Database [18], a seeded decade-stratified draw from argued cases with a clean outcome and public question-and-facts material, these being the first forty-eight in manifest order; each packet holds the question presented, party names, a factual summary, and a lower-court opinion excerpt, the Court’s decision excluded. The model argues an assigned side in under 900 words, and no measurement here predicts a case outcome. A second task uses forty-eight published F.3d decisions from the Caselaw Access Project archive (1995 to 2019, four from each of the First through Eleventh and Federal Circuits, no cross-appeals); the packet is the opinion’s statement of the case and facts, cut before the analysis and scrubbed of outcome sentences, argued for the appellant under all five conditions.

Conditions. The five conditions are baseline; quota (“cite at least eight distinct case authorities”); temporal (“every authority you cite MUST have been decided before 1970”), a clause no user writes, a checkable stand-in for the authority limits practitioners impose; stakes (“this section will be filed with the court; citing authority that does not exist, or misquoting authority, risks sanctions under Rule 11”); and combined, which concatenates the three clauses, so a combined effect cannot be assigned to one of them. Compliance is partial and uneven, 0 to 35 percent of checked citations violating their condition, so contrasts are intention-to-treat across different doses. Each condition has one fixed template, at temperature 0. The stakes clause names Rule 11 [1], which gov-

erns district-court civil actions [19] and neither task’s forum, so it is a stakes signal only. A second template rephrasing every condition was run on all 48 matters. A *grounded* arm reran baseline and combined with the ten U.S. Reports opinions most similar to the question and facts in the prompt (TF-IDF over 25,250 cached opinions decided before argument and, for combined, before 1970, neither party named), each with citation, year, and its best-matching 250-word passage.

Models. The models are Qwen3-30B-A3B, Mistral Small, Llama-4 Maverick, DeepSeek V4 Flash, Grok 4.3, GPT-5.4-mini, and Claude Sonnet 5, one per vendor, run through OpenRouter in August 2026 with completion limits of 1,400 to 12,000 tokens (identifiers in the repository). Tables are ordered by baseline strict quotation accuracy, and Sonnet 5, GPT-5.4-mini, and Grok 4.3, with the highest baseline existence, are the top of the range. An eighth model, GLM-4.7-flash, exhausted a 30,000-token budget in reasoning on 33 of 240 drafts and is excluded.

Adjudication. Full case citations are extracted with eyecite [20] (statutes and rules cite no case and are outside every rate) and labelled exists, not found, or unverifiable, the last for a vendor database number and for a reporter the index carries nothing from, chiefly agency reporters, so coverage is never counted as fabrication (0.6 percent of 14,040 records). Quotations are located by balanced quotation marks, which block quotations lack (12 block-quoted lines in 1,680 drafts, none Sonnet 5’s); a parallel citation string is read as one case and procedural history such as *cert. denied* is not a source; quotations are attributed, in order, to the citation whose parenthetical holds them, the case named in their own sentence, the citation following them in that sentence (not one introduced by *see* or *cf.*), the citation an *Id.* points to, or the last citation in the paragraph; one introduced by a statute, rule, the record, or a lower court, or with no citation in its paragraph, is left unpaired, the checker having nothing to test it against: 37 percent of the 10,846 quotations in the full run: 24 percent introduced by non-case material, 10 percent in a paragraph that cites nothing, 3 percent too far from any citation. *Exists* means the volume, reporter, and page resolve in the 18-million-row index (13,241 of the 14,040 records; 720 are not found); for the 11,769 whose draft names the parties the name is compared separately, and 205 (1.7 percent) carry a name the index does not match; counting every mismatch as not found keeps the 3 surviving existence contrasts, adds Mistral Small’s temporal fall (corrected $p=0.011$), and removes Sonnet 5’s uncorrected combined rise. A quotation q is split into fragments $f_1, \dots, f_n$ at ellipses and bracketed alterations (a lawful substitution is read as an omission and the words around it must match), alteration parentheticals dropped, and compared with the normalized text T of the case it is attributed to: *accurate* when every f_j is a substring of T , *near miss* when $\min_j \text{cov}(f_j, T) \geq 0.85$, for cov the share of a fragment’s words of five or more letters occurring in T , and *inaccurate* otherwise. Matching is case and punctuation insensitive, and a quotation that drops an internal citation without leaving an ellipsis passes when the opinion holds the fragment’s halves in order across a citation-shaped gap; the lenient rate is a diagnostic: 0.85 coverage admits one wrong word in seven. The quotation checker scores 320 of 320 on twenty synthetic drafts planting 16 item kinds, eight of them attribution traps, and the existence and pinpoint checks 100 of 100. Pinpoint pages are verified against the official pagination in the Caselaw Access Project’s HTML volumes, which mark every page break; a pincite naming a range or list is credited on any page it names, a quotation is located by its longest fragment, and only those whose fragment is somewhere in the cited case enter the page-level rate.

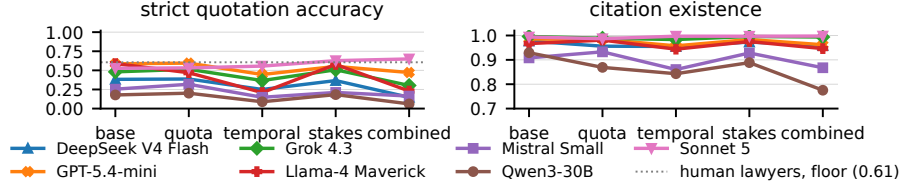


Figure 1. Strict quotation accuracy (left) and reporter-coordinate existence (right) by condition, one line per model; the dotted line is the human reference of Section 3, a severity floor on OCR-converted briefs, not a comparison; the right axis starts at 0.7.

Human reference. The same instrument scored 482 pre-ChatGPT appellate brief excerpts from Liu et al. [3] (every brief-derived excerpt carrying no injected error, their model-written subset excluded; OCR-converted, so a floor); human lawyers reach 0.606 strict and 0.788 lenient quotation accuracy (416 scored) and 0.961 existence; crediting the 59 failures that are near misses without any alteration, the signature of a corrupted character, lifts the floor to 0.748. It is a severity reference only: human briefs differ in citation mix and the remaining failures are unaudited.

Inference. The primary analysis is a citation-level logistic generalized estimating equation per model, $\text{logit}P(y_{imk} = 1) = \beta_0 + \beta_k$ for the existence or accuracy indicator y_{imk} of item i in matter m under condition k , exchangeable within matters [21], so the 48 matters are the independent units. Each model has 8 contrasts (four conditions, two outcomes); we report Holm-corrected p values within model, the family the paper draws conclusions about, and call an effect significant only when it survives; counts of models are descriptive, and the effective sample is 48 matters. A matter-level cluster bootstrap, fixed at freeze (2,000 resamples, on within-matter differences), agrees on seven of the eight survivors; Qwen3-30B’s temporal quotation fall is the one touching zero (CI -0.155 to 0.004, sign kept in 0.966). Every other fit is exploratory and follows the same model and the same rule, Holm within model over that fit’s own tests: the lenient outcome, distinct authorities, the second template, and pooled templates, tasks, and generations (Section 4), each pooling with its fixed effect; the released matrix gives every contrast under every fit. For the revision experiment, pre-registered descriptively, we add matter-paired comparisons between arms (Wilcoxon, bootstrap interval). Hypotheses and decision rules were frozen before the full run; every later decision, including two withdrawn findings, is in a dated ledger.

4. Experiment 1: Pressure

Table 1 and Figure 1 report the rates.

Reporter-coordinate existence. Only the 30B model’s existence moves significantly, 0.929 to 0.775 under the combined condition, with the quota, temporal, and combined contrasts all surviving correction (combined OR 0.29, CI 0.17 to 0.48, corrected $p < 0.001$). Mistral Small falls without reaching significance and the other five stay between 0.94 and 1.00 everywhere, on cells as small as 2 against 1 not found, so those nulls are failures to reject, not evidence of no effect. Each rests on at least 26 citations

Table 1. Baseline versus the combined condition; quotation columns give the strict rate over attributed quotations (lenient) [scored], existence columns the rate [not found / adjudicable], the last the share of checked citations decided after 1969 under the temporal condition; an asterisk marks a contrast surviving Holm correction within model.

Model	Quotes, base	Quotes, comb.	Exist, base	Exist, comb.	Viol., temporal
Qwen3-30B	0.178 (0.609) [230]	0.061 (0.322) [326]*	0.929 [26/368]	0.775 [89/396]*	20%
Mistral	0.254 (0.616) [185]	0.164 (0.455) [244]	0.908 [34/370]	0.868 [69/521]	35%
DeepSeek	0.383 (0.733) [251]	0.147 (0.491) [285]*	0.977 [8/352]	0.959 [20/488]	11%
Grok 4.3	0.480 (0.757) [152]	0.302 (0.647) [119]	0.996 [1/253]	0.994 [3/479]	0%
Sonnet 5	0.527 (0.802) [258]	0.649 (0.820) [194]	0.994 [2/317]	0.998 [1/450]	0%
GPT-5.4-mini	0.586 (0.820) [128]	0.472 (0.697) [89]	0.986 [6/415]	0.962 [27/713]	13%
Llama-4	0.588 (0.784) [51]	0.230 (0.557) [61]*	0.967 [7/211]	0.946 [27/502]	10%

not found per arm, a record being one occurrence of a full citation; refitting on distinct authorities per draft leaves the same 8 contrasts.

Quotation accuracy. The temporal restriction damages quoting most; after correction it falls under temporal and combined pressure for Llama-4 (0.588 to 0.230) and Qwen3-30B (0.178 to 0.061) and under the combined condition for DeepSeek; the other three fall in the same direction without surviving correction, and Sonnet 5 does not fall (below). Llama-4 shows the largest drop in this fit, 0.588 to 0.209 under the temporal restriction (OR 0.18, CI 0.08 to 0.42, corrected $p < 0.001$) on cells of 51 and 43, and it does not replicate under the second template; on the lenient outcome 7 contrasts survive. A fall under the date clause could be damage to quoting or a change in what is quoted, since it moves the median year of Sonnet 5’s cited cases from 1986 to 1943. The clause sets that year, so it is a mediator: adjusting for it decomposes the effect rather than removing a bias, and odds ratios are not collapsible, so part of the movement is mechanical. Pooled over models with a model effect on the 3,655 dated quotations, the temporal clause’s odds ratio moves from 0.62 to 0.77 ($p=0.033$) once decision year enters the fit and the combined condition’s from 0.56 to 0.68 (CI 0.53 to 0.88, $p=0.003$), removing 45 and 35 percent of the log-odds effect respectively. The assumption-light alternative disagrees: inside a pre-1970 bucket, which still leaves the arms twelve years apart in median, neither pooled contrast survives correction (combined Holm 0.094). With three generations pooled, the combined-condition fall survives correction for 6 of the seven models (OR 0.21 to 0.54), Sonnet 5 excepted, and Mistral Small’s existence fall also survives (OR 0.59, corrected $p=<0.001$).

Accurate quotations per draft. The rate R_d is conditional on quoting: scored quotations per cell run from 43 to 326 and the unpaired share from 15 to 50 percent, moving with the treatment. We therefore also test, post hoc and on both prompt templates, the count K_d (Wilcoxon paired by matter, $n=48$, Holm within model). On the first template the count falls numerically from baseline to combined for all seven models and survives correction for 8 contrasts in five models, all under the temporal or combined clause: Qwen3-30B, DeepSeek (2.00 to 0.88 per draft), Llama-4 (0.62 to 0.29), and two whose rate contrasts did not survive, GPT-5.4-mini (corrected $p=0.036$) and Grok 4.3 ($p=0.011$). The second template replicates the direction: the contrast keeps its sign in 20 of the 26 tests that have one, six of the eight first-template survivors fall again, and six of the seven lose correct quotations on at least one template, though which model is caught moves with the wording. Sonnet 5’s count is flat (2.83 to 2.62, $p=0.579$) while its

failing quotations per draft, near misses and inaccurate together, fall from 2.54 to 1.42, the evidence for selection below.

Robustness to phrasing. Under the second template 13 contrasts survive, 5 of them among the 8 above: Qwen3-30B’s four and DeepSeek’s combined quotation fall, so Llama-4’s two quotation falls and Qwen3-30B’s quota existence fall are single-template results. Pooling both, all 8 survive and the combined quotation fall survives for 6 of seven models, Sonnet 5 excepted.

Second task and retrieval. Baseline existence on the 48 appellate matters runs from 0.652 for Qwen3-30B to 0.993 for Sonnet 5, lower than on the Supreme Court matters for 7 of seven, and concentrates in the federal reporters, 22 percent against 4 percent for the U.S. Reports; the index resolves all but 3 of the 759 federal-reporter citations in the deciding opinions themselves, so the shortfall is not coverage. A grounded arm with ten verified authorities in the prompt cut citations not found from 320 in 5,835 to 55 in 5,586 and raised quotation accuracy in every cell, yet no cell exceeded 0.753 with the source passages there. One pressure contrast survives within the task, Llama-4’s existence fall under the temporal clause (corrected $p=0.031$). Pooling both tasks, 7 of the 8 survivors remain, joined by 5 further falls and Sonnet 5’s rate rise under the stakes clause (OR 1.62, corrected $p=0.007$).

Sonnet 5 does not degrade; the leading explanation is that it quotes less. Sonnet 5’s strict rate is the highest of the seven under the combined condition and rises (0.527 to 0.649; uncorrected $p = 0.0251$, corrected $p = 0.201$), a rise that survives no fit. It wrote 9.6 quotations per draft at baseline and 6.1 under the combined condition, so the quotations it stopped writing were disproportionately the ones that would have failed. Selection is the leading reading; which were dropped, by length or case age, is not tested. With three generations pooled Sonnet 5’s existence rise survives the logistic fit (corrected $p=0.152$) but rests on twelve citations not found (4 of 972 against 3 of 1,322), on which an exact test gives $p=0.47$; the quotation rise reaches corrected $p=0.198$ there. In the appellate task the stakes clause alone raises the rate (0.468 to 0.624, corrected $p=0.049$) without moving the cited cases (median year 1991 in both), so the direction holds across tasks.

5. Experiment 2: The Checker inside the Revision Loop

Combined-condition drafts entered a revision loop: each round the checker returned one line per failure, the model revised, and the loop ended on a clean draft or after three revisions (a round is a draft the checker read). A *false feedback* control gave the same number of identical lines aimed at correct items, *mixed* controls replaced each line with a false one at random so that the feedback’s nominal precision π , the share of its lines drawn from the checker, was 0.25, 0.5, or 0.75 ($\pi = 1$ under true and $\pi = 0$ under false feedback), replacing rather than adding lines, so recall falls with precision and the arms vary both. A *no-feedback* control gave the same rounds with a generic instruction to revise. The loop used the first 24 matters; rates are means over drafts with at least one scored quotation (68 of the 168 true-arm episodes ended with none, two without passing the checker, 19 of Grok 4.3’s 24, so its final rate rests on 5 drafts and Llama-4’s on 6). The loop was rerun after the attribution rule was corrected; the original run, whose true-arm flags had realized precision 0.71 under the corrected rule (0.38 for Sonnet 5),

Table 2. Revision under true checker feedback, 24 episodes per model. Round-0 quotations, flagged (inaccurate or near miss) and accurate, traced to the final draft. corr.: an accurate quotation on the same citation stands in its place; left: still flagged; removed: deleted, dequoted, or made inaccurate; added: new, accurate/flagged; strict accuracy as the draft mean, pooled in parentheses, round 0 / final; Clean: passing draft, with quotations / with none.

Model	Flagged at round 0					Accurate at round 0			Strict, r0 → final (pooled)	Clean
	n	corr.	dequoted	deleted	left	n	removed	added		
Qwen3-30B	139	0	10	13	116	12	0	0/32	0.089 → 0.092 (0.079/0.075)	0/0
Mistral	92	4	4	40	44	14	1	1/27	0.238 → 0.354 (0.132/0.202)	3/7
DeepSeek	134	2	58	71	3	31	5	1/2	0.301 → 0.958 (0.188/0.853)	11/12
Grok 4.3	37	0	23	14	0	13	3	0/0	0.234 → 1.000 (0.260/1.000)	5/19
Sonnet 5	35	3	2	30	0	69	4	0/0	0.743 → 1.000 (0.663/1.000)	23/1
GPT-5.4-mini	22	1	1	19	1	31	5	0/1	0.538 → 0.857 (0.585/0.931)	11/9
Llama-4	33	0	5	28	0	11	2	0/0	0.273 → 1.000 (0.250/1.000)	6/18

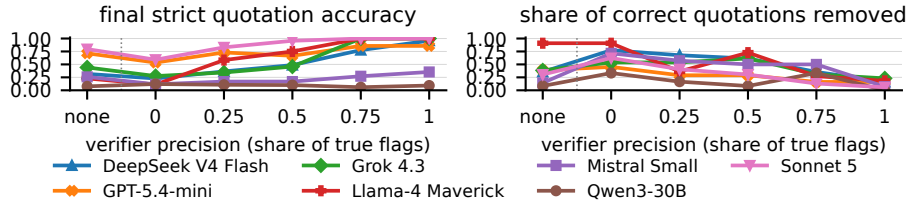


Figure 2. Nominal verifier precision π (share of true feedback lines; leftmost point, no feedback) against final strict quotation accuracy (left) and the share of accurate round-0 quotations that revision deleted, dequoted, or made inaccurate (right).

is released beside it. The rerun’s flags are the corrected rule’s verdicts, whose precision on inaccurate verdicts the two readings of Section 6 bracket between 22 and 35 percent wrong; Figure 2’s π is therefore nominal, its arm at $\pi = 1$ realising 65 to 78 percent. Every scored round-0 quotation was traced to the final draft (Table 2).

Satisfying the checker by deletion. Under true feedback the final strict rate runs from 0.092 for Qwen3-30B to 1.000, which Sonnet 5, Grok 4.3, and Llama-4 all reach (Table 2). Sonnet 5 had 35 flagged quotations at round 0: 1 corrected in place, 2 replaced by an accurate quotation from the same case, 2 dequoted, 30 deleted, while only 4 of its 69 accurate quotations went, against 21 under revision with no feedback. Qwen3-30B left most flags standing and Mistral Small nearly half; the other five removed most of what was flagged. Dequoting is not deletion: the words stay, the citation stays in 100 of 103 cases, and dropping the marks is the right repair for a paraphrase. Counting it as repair, deletion still exceeds it for 6 of seven models, Grok 4.3 excepted, though the comparison bears only on the flags a model acted on: Qwen3-30B acted on 23 of 139 and left the rest. Counted over all 24 episodes, accurate quotations per draft rose for 1 of seven models (Mistral Small, 0.58 to 0.75) and were flat or lower for the rest (Sonnet 5 2.88 to 2.83), so the loop raised the rate over surviving quotations and not the number of correct ones in a draft. Existence rose by the same route: of 108 citations unresolved at round 0, 78 were gone from the final draft. A *passage* arm carried the closest passage of the cited opinion in each quotation line. Repair rose in six of seven models, Grok 4.3

excepted, to 13 of 35 for Sonnet 5 and 27 of 92 for Mistral Small against 4, and deletion alone still exceeded repair and dequoting together in 4 of seven.

Controls: feedback content and verifier precision. Without feedback (Figure 2) Sonnet 5’s strict rate still rose (0.743 to 0.796), and true feedback added 0.20 over revision alone (paired over the 23 matters scored in both arms; interval 0.11 to 0.31, $p=0.003$), as for DeepSeek (0.39, $n=12$); the differences for the other five are within noise. Sonnet 5’s correct quotations removed rise as precision falls, 4, 9, 21, 28, and 43 of 69 at nominal π of 1, 0.75, 0.5, 0.25, and 0, against 21 with no feedback. Realised precision is below the label, the two hand readings implying flag precision between 65 and 78 percent, and its accurate side was audited at 0 of 40 wrong. The loop stops on a clean draft, so the arms differ in rounds executed, 2.47 under true feedback against 3.21 under false, and the series is monotone in both. Holding rounds fixed keeps it: removals per episode-round run 0.048, 0.090, 0.145, 0.150, 0.200 against 0.109 with no feedback, and after one round 10 to 32 percent of the 181 accurate quotations are gone against 23. No feedback sits between π of 0.75 and 0.5 on both bases, interpolating to 0.56 and 0.66, or 0.37 to 0.52 realised, lower bounds since the audited range understates the current rule. Below it a verifier destroys more correct work than revision alone; the arms move recall with precision, so this bounds a loop of this shape, not precision alone. Under all-false feedback five of seven models ended below their starting rate while flagged counts fell in every model: what changed was the deletion of correct quotations.

6. Residual Errors at the Top of the Range

Paraphrase inside quotation marks. Decomposing every inaccurate quotation against the cited opinion’s text, 68 percent of Sonnet 5’s (169 of 249) are paraphrases of the correct case presented as verbatim quotation, 56 match nothing substantial in it, and 24 cite authority outside the U.S. Reports.

Real language, wrong case. Searching every inaccurate quotation against the 62,049 U.S. Reports opinions in the Caselaw Access Project cache locates 319 whose language is verbatim in another opinion (135 with the true source cited in the same draft, so a ceiling). Sonnet 5’s 249 inaccurate quotations partition by provenance into 73 verbatim elsewhere, 130 found nowhere, 40 stock phrases, 6 present in the attributed case after all. Two independent readings of the same 160 inaccurate verdicts put 22 and 35 percent on the checker’s side, non-case material in quotation marks or a quotation bound to the wrong case, agreeing on 9 at a kappa of -0.10: neither share is reliable, and between a fifth and a third of these verdicts are the instrument’s. The contrasts are refit against the pessimistic bound: counting an item as the checker’s whenever either reader did, 51 percent of the sample, and reclassifying inaccurate verdicts as unpaired at that rate per condition (200 draws), 7 of the 8 survivors hold in every draw, all but Qwen3-30B’s temporal quotation fall. The count K_d is insulated, since reclassifying a verdict as unpaired leaves the accurate count untouched; the year-adjusted fit is not. The accurate side is checked the same way: of the 1,341 accurate quotations running six words or more and present in the archive, 1,267 lead back to the case the draft attributes them to; the rest of the 2,169 are short quotations or citations to reporters the archive does not carry. The 74 that do not trace are mostly those same outside reporters (30) or language common to several opinions; the check bounds attribution error rather than estimating it, since it cannot separate a miss from a gap in the archive.

Right case, wrong opinion. The archives hold opinion, concurrences, dissents, and head matter in one record, so a dissent’s words score accurate against the case. Of the 1,599 accurate quotations the archive’s markers place (74 percent of those scored accurate), 39 sit outside the Court’s opinion (20 concurrence, 11 dissent, 8 syllabus), 33 of them carrying no signal in the draft, 2.1 percent of the placed.

Right case, wrong page. Of the 9,129 pinpoints, 92 percent were checked against the cited case’s page span (the rest cite other reporters or cases past volume 572) and 1,295 carry a quotation the case contains somewhere, the page-level rate’s scope; for those (75 to 305 per model) the quotation is on a cited page 85 percent of the time for Sonnet 5, 55 to 69 for the other six, 85 for the briefs, and on an adjacent page 10 percent.

7. Discussion

Where the checker belongs. An inference, since no arm had a person in it: prefer the checker at the gate to a loop resolving bare flags. Removing an unverifiable quotation answers a flag defensibly, but false flags get the same answer and the tool still reports a clean draft.

Verifier precision matters. Our checker’s error is 22 to 35 percent by the two hand readings, and published detectors span this axis: the highest-recall agent reaches 84.4 percent recall at 40.8 percent precision, the highest-precision one 76.1 at 62.8 percent recall [3], the first below the crossing this loop gives and the second above it. A pipeline feeding a detector’s flags to a model should report precision beside recall.

What evaluations should include. The observed rate is $R_d = K_d/|S_d|$, and a model can raise it by shrinking what it exposes, $|S_d|$, as well as by raising correct output K_d : under pressure Sonnet 5 quoted less, under feedback models deleted what was flagged, $|S_d|$ down and K_d flat or lower in both. An evaluation scoring only what a system emits should report K_d beside R_d .

8. Limitations

Both tasks are United States federal case law and the stakes clause names a rule governing neither forum; the checks stop at existence, wording, and page: a relevance panel fell short of a 0.75 bar set in advance from reported judge agreement [22], so whether a quotation supports its proposition, or the case is good law, is untested. All of it after the primary fit is exploratory. The two readings behind the checker’s error cover inaccurate verdicts only, neither was blind to model or condition, and they agree no better than chance, so we report the disagreement and test against the pessimistic bound rather than a point estimate. The between-template spread (16 points) exceeds many single-run effects, the models are one per vendor, and the range ordering rests on baseline existence.

9. Conclusion

No one failure mode covers the seven: reporter-coordinate existence falls, in the primary fit, only for the 30B model, quotation accuracy for three, and correct quotations per draft for six across the two templates, two of them at the top of the range, where the rate does not move. Acting on its flags, a model satisfies the checker by removal.

References

- [1] *Mata v. Avianca, Inc.*, 678 F. Supp. 3d 443 (S.D.N.Y. 2023); 2023. Opinion and order on sanctions, No. 1:22-cv-01461 (PKC), June 22, 2023.
- [2] Charlotin D. AI Hallucination Cases Database; 2025. Accessed 2 September 2026; 2,008 cases listed. <https://www.damiencharlotin.com/hallucinations/>.
- [3] Liu P, Stambach D, Henderson P. Who Checks the Citations? Benchmarking Legal Hallucination Detection; 2026. Accepted at the Conference on Language Modeling (COLM) 2026. Available from: <https://arxiv.org/abs/2606.21155>. doi:10.48550/arXiv.2606.21155.
- [4] Chen S, Yin H, Zhou S. LegalCiteBench: Evaluating Citation Reliability in Legal Language Models; 2026. Available from: <https://arxiv.org/abs/2605.10186>. doi:10.48550/arXiv.2605.10186.
- [5] Zhao C, Tang Y, Qian Y. Do Deployment Constraints Make LLMs Hallucinate Citations? An Empirical Study across Four Models and Five Prompting Regimes; 2026. Available from: <https://arxiv.org/abs/2603.07287>. doi:10.48550/arXiv.2603.07287.
- [6] Harvard Law School Library Innovation Lab. Caselaw Access Project; 2024. Digitized collection of official, book-published U.S. case law; accessed September 2026. <https://case.law/>.
- [7] Dahl M, Magesh V, Suzgun M, Ho DE. Large Legal Fictions: Profiling Legal Hallucinations in Large Language Models. *Journal of Legal Analysis*. 2024;16(1):64-93. doi:10.1093/jla/laae003.
- [8] Magesh V, Surani F, Dahl M, Suzgun M, Manning CD, Ho DE. Hallucination-Free? Assessing the Reliability of Leading AI Legal Research Tools. *Journal of Empirical Legal Studies*. 2025;22(2):216-42. doi:10.1111/jels.12413.
- [9] Gehring J, Zheng K, Copet J, Mella V, Cohen T, Synnaeve G. RLEF: Grounding Code LLMs in Execution Feedback with Reinforcement Learning. In: *Proceedings of the 42nd International Conference on Machine Learning*. vol. 267 of *Proceedings of Machine Learning Research*. PMLR; 2025. p. 19034-55. Available from: <https://proceedings.mlr.press/v267/gehring25a.html>.
- [10] Blair-Stanek A, Van Durme B. LLMs Provide Unstable Answers to Legal Questions. In: *Proceedings of the Twentieth International Conference on Artificial Intelligence and Law (ICAIL 2025)*. Chicago, IL, USA: ACM; 2025. p. 425-9. doi:10.1145/3769126.3769245.
- [11] Gultekin S, Rossi Reich M, Galloni F, Lagioia F, Consiglio E, Sartor G, et al. A Two-Dimensional Evaluation Framework for Factual and Reasoning Assessment of LLMs in Legal Question Answering. In: *Legal Knowledge and Information Systems: JURIX 2025*. vol. 416 of *Frontiers in Artificial Intelligence and Applications*. IOS Press; 2025. p. 86-97. doi:10.3233/FAIA251579.
- [12] Sanchi M, Novotná T. Towards Hybrid Evaluation Methodologies for Large Language Models in the Legal Domain. In: *Legal Knowledge and Information Systems: JURIX 2024*. vol. 395 of *Frontiers in Artificial Intelligence and Applications*. IOS Press; 2024. p. 389-92. doi:10.3233/FAIA241279.
- [13] Garneau N, Palmer Olsen H, Corduant A, Tarissan F. Combining Network and Text to Provide Legal Pincites. In: *Legal Knowledge and Information Systems: JURIX 2024*. vol. 395 of *Frontiers in Artificial Intelligence and Applications*. IOS Press; 2024. p. 282-7. doi:10.3233/FAIA241256.
- [14] Zhang L, Savelka J, Ashley KD. Do LLMs Truly “Understand” when a Precedent Is Overruled? In: *Legal Knowledge and Information Systems: JURIX 2025*. vol. 416 of *Frontiers in Artificial Intelligence and Applications*. IOS Press; 2025. p. 239-50. doi:10.3233/FAIA251592.
- [15] Nguyen HT, Satoh K. ConsRAG: Minimize LLM Hallucinations in the Legal Domain. In: *Legal Knowledge and Information Systems: JURIX 2024*. vol. 395 of *Frontiers in Artificial Intelligence and Applications*. IOS Press; 2024. p. 327-32. doi:10.3233/FAIA241263.
- [16] Gray MA, Savelka J, Oliver WM, Ashley KD. Generating Legal Arguments with Automatically Identified Factor Magnitudes. In: *Proceedings of the Twentieth International Conference on Artificial Intelligence and Law (ICAIL 2025)*. ACM; 2025. p. 199-208. doi:10.1145/3769126.3769212.
- [17] Pires R, Malaquias Junior R, Nogueira R. Automatic Legal Writing Evaluation of LLMs. In: *Proceedings of the Twentieth International Conference on Artificial Intelligence and Law (ICAIL 2025)*. ACM; 2025. p. 420-4. doi:10.1145/3769126.3769227.
- [18] Spaeth HJ, Epstein L, Nelson MJ, Martin AD, Segal JA, Ruger TJ, et al. Supreme Court Database, Version 2025 Release 01; 2025. Modern database, terms 1946 to 2024. <https://scdb.la.psu.edu/>.
- [19] Federal Rules of Civil Procedure, Rule 11 (Signing Pleadings; Sanctions) and Rule 1 (Scope and Purpose); 2024. Rule 11(b)–(c) states the certification and sanctions; Rule 1 limits the rules to “all civil actions and proceedings in the United States district courts, except as stated in Rule 81”. https://www.law.cornell.edu/rules/frcp/rule_11.

- [20] Cushman J, Dahl M, Lissner M. eyecite: A tool for parsing legal citations. *Journal of Open Source Software*. 2021;6(66):3617. Available from: <https://doi.org/10.21105/joss.03617>. doi:10.21105/joss.03617.
- [21] Liang KY, Zeger SL. Longitudinal data analysis using generalized linear models. *Biometrika*. 1986;73(1):13-22. doi:10.1093/biomet/73.1.13.
- [22] Zheng L, Chiang WL, Sheng Y, Zhuang S, Wu Z, Zhuang Y, et al. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena; 2023. *NeurIPS 2023 Datasets and Benchmarks Track*. Available from: <https://arxiv.org/abs/2306.05685>. doi:10.48550/arXiv.2306.05685.